

Dear Patient, your metabolic risk assessment has been processed by the AgentXHealth multi-agent clinical AI system. This report combines biochemical, physical, and demographic data to estimate your Type 2 Diabetes risk profile.

Overall Assessment Result

Risk Score: 60%

HIGH RISK



0%50%100%

Ensemble probability score: 60%

PHYSICIAN'S AI SUMMARY

Fasting glucose of 110 mg/dL is in the prediabetes range (100-125 mg/dL). Without intervention, prediabetes carries a significant risk of progression to Type 2 diabetes within 5-10 years. Lifestyle modifications including dietary changes and increased physical activity can substantially reduce this risk.

URGENT CLINICAL ALERTS

- Low blood pressure (BP <= 90 mm Hg). Monitor for symptoms.

COMPUTED METABOLIC INDICES

HOMA-IR and QUICKI are physiologically validated indices that quantify insulin resistance beyond raw glucose and insulin values alone.

Index	Your Value	Clinical Interpretation
HOMA-IR (Glucose x Insulin) / 405	4.074	Insulin resistance indicated (≥ 2.5) Normal: < 2.5 Resistance: ≥ 2.5

QUICKI 1 / (log(Insulin) + log(Glucose))	0.135	Reduced insulin sensitivity (≤ 0.33) Normal: > 0.33 Reduced: ≤ 0.33
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YOUR CLINICAL METRICS AND AI RISK IMPACT

Each metric is evaluated against established clinical thresholds. AI Risk Impact reflects how each value contributes to your overall risk estimate.

Clinical Metric	Your Result	AI Risk Impact
Fasting Glucose	110 mg/dL	BORDERLINE Prediabetes range (100-125 mg/dL). Lifestyle changes recommended.
Insulin	15 uU/mL	NORMAL Normal fasting insulin (5-25 uU/mL).
Body Mass Index (BMI)	27	BORDERLINE Overweight (BMI 25-29.9). Associated with insulin resistance.
Blood Pressure	80 mm Hg	NORMAL Normal blood pressure (< 120 mm Hg).
Age	42 yrs	BORDERLINE Early midlife (30-44 yrs). Risk begins to rise gradually.
Pregnancies	2	NORMAL 1-2 pregnancies. Minimal additional risk.

RECOMMENDED NEXT STEPS

- Urgent Consult:** Contact your physician or endocrinologist within 3-5 days. Do not delay.

- Comprehensive Metabolic Panel:** Full panel including HbA1c, insulin, lipids, and kidney function.

- Do Not Self-Medicate:** Do not start medications or drastic diets without consulting your doctor.

- **Monitor Symptoms:** Watch for excessive thirst, frequent urination, blurred vision, or fatigue.

- **Emergency Signs:** Confusion, rapid breathing, or loss of consciousness — call emergency services immediately.

About This Report

Generated by AgentXHealth — a multi-agent AI framework using three physiologically specialized models: **Laboratory Agent** (biochemical markers, AUC 0.82), **Physical Agent** (anthropometric indicators), and **Demographic Agent** (age and parity). Outputs are combined via weighted probabilistic fusion (Lab $\times 0.5$, Physical $\times 0.3$, Demographic $\times 0.2$) and refined through clinical rule adjustment for edge-case safety.

Disclaimer: AgentXHealth is an investigational AI clinical decision support tool. This report does not constitute a formal medical diagnosis and is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider before making any medical decisions. | Report ID:

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